

Notes: Baghai & Becker (JFE, 2018)

Non-rating revenue and conflicts of interest

Contents

1	Big picture	2
2	Data and measurement	2
2.1	Institutional setting: Indian rating agencies	2
2.2	Regulatory disclosure shock	2
2.3	Ratings data and rating scale	3
2.4	Main variables	3
2.5	Default sample	3
3	Models and empirical specifications	3
3.1	Ratings specification with issuer, agency, and year fixed effects	3
3.2	Main within issuer-year ratings specification	3
3.3	Default specification	4
3.4	Why the sign matters	4
4	Identification and empirical design	4
4.1	What identifies the ratings effect	4
4.2	Why defaults matter for interpretation	4
4.3	Why rating thresholds matter	4
4.4	Main remaining limitations	5
5	Tables and figures	5
5.1	Table 1, Table 2, and Figure 1: sample, defaults, and raw rating distribution	5
5.2	Figure 2, Table 3, and Table 4: main rating inflation tests	7
5.3	Table 5 and Table 6: lagged payments and threshold incentives	8
5.4	Table 7 and Table 8: robustness to financial-sector issuers and ordered probit	9
5.5	Figure 3, Figure 4, Table 9, and Table 10: default evidence	10
5.6	Table 11 and Table 12: fee amounts and default rates	12
6	Short summary	12
7	Conclusion	13
7.1	Core conclusion	13
7.2	Economic intuition	13
7.3	Takeaway	13

1 Big picture

- **Question.** Do rating agencies give more favorable ratings to issuers that pay them for non-rating services, and are those favorable ratings justified by lower default risk?
- **Conflict.** Rating agencies sell ratings used by investors, but most of their revenue comes from issuers. Non-rating services deepen this commercial relationship and may create a stronger incentive to please the issuer.
- **Setting.** The paper uses Indian rating-agency disclosures after a 2010 SEBI rule requiring information on non-rating income from rated issuers.
- **Data advantage.** The authors observe whether an issuer pays a particular rating agency for non-rating services and, for most observations, the amount paid.
- **Main identifying comparison.** Many Indian issuers are rated by more than one agency. The key tests compare ratings assigned to the same issuer in the same year by agencies with versus without a non-rating-services relationship.
- **Main ratings result.** Agencies that provide non-rating services rate the same issuer about 0.3 notches better. When fees are measured continuously, a doubling of fees is associated with about a 0.4-notch better rating.
- **Threshold result.** The effect is much larger near economically important ratings thresholds such as AA- and BBB-. In these subsamples, estimates are roughly one notch or more.
- **Default result.** Within a rating category, issuers that pay for non-rating services default more often. This is important because a better rating would be warranted only if those issuers had lower or similar default risk.
- **Economic takeaway.** The paper provides direct evidence that commercial ties between raters and issuers are associated with both rating inflation and lower rating quality.

2 Data and measurement

2.1 Institutional setting: Indian rating agencies

- The sample covers Indian corporate issuers and rating agencies during 2010–2015.
- The main rating agencies in the data are CRISIL, ICRA, CARE, Brickwork Ratings, and INDRA.
- Several Indian agencies are linked to global rating agencies: S&P owns CRISIL, Moody’s owns a large stake in ICRA, and INDRA is a Fitch subsidiary.
- Indian rating agencies provide both ratings and non-rating services, including risk-management services, industry analysis, business analytics, IT/business process services, and management consulting.
- Firewalls formally separate ratings and non-ratings businesses, but the paper asks whether the commercial relationship still affects ratings.

Interpretation. The setting is useful because the Indian regulator made issuer-level non-rating income visible. Most other markets do not provide this type of direct payment data.

2.2 Regulatory disclosure shock

- In May 2010, SEBI Circular CIR/MIRSD/CRA/6/2010 required rating agencies to disclose annual revenue information.
- The key disclosure requires the issuer-wise percentage share of non-rating income relative to total revenue from that issuer.
- Some agencies also voluntarily disclosed enough information to recover the level of non-rating revenue paid by an issuer.
- These disclosures allow the authors to construct both an extensive-margin variable and an intensive-margin fee variable.

Interpretation. The empirical test is possible because the paper observes payments at the issuer-agency level. This is exactly the level at which conflicts of interest operate.

2.3 Ratings data and rating scale

- Ratings and firm information come from CMIE Prowess.
- Ratings are reported at the debt-instrument level and then averaged to the issuer-agency-year level.
- The numerical rating scale is ordered so that lower numbers are better: 1 = AAA and 19 = C-.
- The main sample has 26,760 issuer-agency-year observations and 7,083 firms.
- Non-rating services are rare in the full sample: 1,165 observations, or 4.35%, involve non-rating services.

Interpretation. Because a lower number means a better rating, a negative regression coefficient means the agency is assigning a more favorable rating.

2.4 Main variables

- **Issuer rating:** average numerical rating assigned to issuer i by agency j in year t .
- **Non-rating services:** indicator equal to one if issuer i pays agency j for non-rating services in year t .
- **Ln(Non-rating issuer revenue):** $\log(1 + \text{non-rating revenue})$, where revenue is measured in million Rupees.
- **Default in $t + 1$:** indicator equal to one if issuer i defaults on any debt instrument in year $t + 1$.

Interpretation. The dummy captures whether there is a consulting/commercial tie. The log revenue measure captures whether larger payments generate stronger effects.

2.5 Default sample

- The default tests use 2010–2014 observations because default information is available through September 2015.
- The default sample contains 18,707 issuer-agency-year observations and 706 defaults.
- The average one-year default rate is 3.8%.
- Investment-grade issuers have an average default rate of 1.3%; high-yield issuers have an average default rate of 8.3%.

Interpretation. Defaults are the key quality check. If consulting clients truly deserve better ratings, then their default rates within rating categories should not be higher.

3 Models and empirical specifications

3.1 Ratings specification with issuer, agency, and year fixed effects

The first ratings model is

$$\text{Issuer Rating}_{i,j,t} = \alpha X_{i,j,t} + \beta_j + \gamma_i + \delta_t + \varepsilon_{i,j,t}, \quad (1)$$

where i indexes issuers, j indexes rating agencies, and t indexes years. $X_{i,j,t}$ is either *Non-rating services* or *Ln(Non-rating issuer revenue)*.

Interpretation. This specification compares ratings after controlling for issuer, agency, and year differences. It addresses cross-sectional selection into consulting but does not fully address time-varying issuer quality.

3.2 Main within issuer-year ratings specification

The main specification is

$$\text{Issuer Rating}_{i,j,t} = \alpha X_{i,j,t} + \beta_{j \times t} + \gamma_{i \times t} + \varepsilon_{i,j,t}. \quad (2)$$

Interpretation. The issuer-year fixed effect $\gamma_{i \times t}$ compares agencies rating the same issuer in the same year. The agency-year fixed effect $\beta_{j \times t}$ absorbs time-varying agency strictness. This is the paper’s strongest ratings design.

3.3 Default specification

For default tests, the paper estimates linear probability models of the form

$$\text{Default in } t + 1_{i,t} = \alpha X_{i,j,t} + \Psi_{i,j,t} + \varepsilon_{i,j,t}, \quad (3)$$

where $X_{i,j,t}$ is a non-rating revenue measure and $\Psi_{i,j,t}$ includes rating controls and fixed effects.

Interpretation. The default tests ask whether, holding the rating category fixed, consulting clients default more. A positive coefficient means the rating is too optimistic relative to realized default risk.

3.4 Why the sign matters

- Ratings regression: negative coefficient $\Rightarrow$ better rating, because 1 is AAA and 19 is C-.
- Default regression: positive coefficient $\Rightarrow$ higher default probability.

Interpretation. The main conflict-of-interest pattern is therefore: negative coefficient in the ratings tables plus positive coefficient in the default tables.

4 Identification and empirical design

4.1 What identifies the ratings effect

- The strongest ratings tests use issuers with multiple raters.
- The comparison is within issuer-year: agency A and agency B rate the same issuer in the same year.
- If agency A receives non-rating revenue from the issuer and agency B does not, the model asks whether agency A gives a better rating.

Interpretation. This design removes all issuer-year fundamentals, including credit quality, leverage, size, profitability, and time-varying issuer shocks.

4.2 Why defaults matter for interpretation

- A consulting relationship could in principle reveal information that helps the rater assign a more accurate rating.
- If so, consulting clients should have lower or similar default risk within a rating class.
- The paper finds the opposite: consulting clients default more within rating categories.

Interpretation. Defaults distinguish favorable treatment from valid information production. Higher default rates are hard to reconcile with better-informed ratings.

4.3 Why rating thresholds matter

- Ratings thresholds such as AA- and BBB- matter for regulation and contracting.
- Issuers near thresholds have especially strong incentives to obtain a small rating improvement.
- The paper finds larger effects near these thresholds and among issuers with split ratings.

Interpretation. Threshold heterogeneity fits the incentives story: rating inflation is most valuable where a small notch change changes investor mandates, borrowing costs, or regulatory treatment.

4.4 Main remaining limitations

- The purchase of non-rating services is not randomly assigned.
- The data capture disclosed contemporaneous and lagged payments, but long-term relationships may matter in ways not fully observed.
- The paper cannot distinguish whether agencies sell favorable ratings or whether issuers use consulting to learn how to game ratings models.
- The evidence comes from India, though the institutional issue is relevant to issuer-pays ratings more generally.

Interpretation. The causal design is not an experiment, but the combination of within-issuer-year comparisons, threshold evidence, and higher default rates makes the conflict-of-interest interpretation compelling.

5 Tables and figures

5.1 Table 1, Table 2, and Figure 1: sample, defaults, and raw rating distribution

Read:

- Table 1 shows that only 4.35% of issuer-agency-year observations involve non-rating services.
- CRISIL has the highest observed non-rating-services incidence among the main agencies: 7.87% of CRISIL observations.
- Firms with better ratings are more likely to use non-rating services: 339 of 1,100 AAA observations involve non-rating services, compared with only 4 of 305 C observations.
- Table 2 shows that defaults are concentrated in lower rating categories. There are no defaults in AAA and AA categories in the default sample.
- Figure 1 shows that issuers with non-rating-services relationships have a distribution shifted toward better numerical ratings.

Economic message: The raw data already show the pattern: non-rating clients tend to receive better ratings. The rest of the paper asks whether this is merely selection or evidence of favorable treatment.

Table 1

Summary statistics—main sample.

Panel A reports a frequency distribution of firm-agency-years with non-rating services, as well as a breakdown by rating agency. The sample spans the years 2010–2015. Panel B tabulates the incidence of firms with multiple ratings in our sample, reported separately for firms that purchase non-rating services and those that do not. Panel C tabulates by rating category observations associated with the purchase of non-rating services and those that are not. Panel D reports the mean, standard deviation, minimum, and maximum of the variables used in the study of ratings. *Issuer rating* is the rating an issuer receives from a given rating agency in a given year, with a value of one denoting the highest credit rating “AAA” and the value 19 denoting “C-.” *Non-rating services* is a dummy variable that takes the value of one if an issuer pays for non-rating services from a given rating agency in a given year, zero otherwise. $\ln(\text{Non-rating issuer revenue})$ is the natural logarithm of one plus non-rating revenue (in million Rupees) paid by an issuer to the rating agency. The variables are defined in more detail in [Section 3](#) of the paper and in Table A-1 of the Appendix.

Panel A			
Full sample			
	Frequency	Percent	
Non-rating services = 0	25,595	95.65	
Non-rating services = 1	1165	4.35	
Total	26,760	100	
CRISIL only			
	Frequency	Percent	
Non-rating services = 0	9299	92.13	
Non-rating services = 1	794	7.87	
Total	10,093	100	
ICRA only			
	Frequency	Percent	
Non-rating services = 0	7962	96.26	
Non-rating services = 1	309	3.74	
Total	8271	100	
CARE only			
	Frequency	Percent	
Non-rating services = 0	5071	98.79	
Non-rating services = 1	62	1.21	
Total	5133	100	
Brickwork Ratings only			
	Frequency	Percent	
Non-rating services = 0	852	100	
INDRA only			
	Frequency	Percent	
Non-rating services = 0	2411	100	
Panel B			
Non-rating services			
Number of raters	No	Yes	Total
1	21,016	603	21,619
2	3809	307	4116
3	577	182	759
4	169	67	236
5	24	6	30
Total	25,595	1165	26,760

Table 1

Continued.

continued.

Panel C					
Rating category	Non-rating services			Total	
	No	Yes			
AAA	761	339		1100	
AA	2468	417		2885	
A	4111	194		4305	
BBB	8336	155		8491	
BB	6818	44		6862	
B	2800	12		2812	
C	301	4		305	
Total	25,595	1165		26,760	

Panel D					
	Obs.	Mean	Std. dev.	Min.	Max.
Issuer rating	26,760	9.044	3.838	1.000	19.000
Non-rating services	26,760	0.044	0.204	0.000	1.000
Ln(Non-rating issuer revenue)	26,468	0.020	0.173	0.000	3.792

Table 2

Summary statistics–default sample.

This table shows summary statistics for the variables used in the analysis of default rates in Section 4.3. The sample spans the years 2010–2014. Panel A reports the number of defaults by rating category. Panel B reports the mean, standard deviation, minimum, and maximum of the variables used in the study of defaults. *Default in $t+1$* is defined at the firm-year level and takes the value of one in year t if a given company has a debt instrument on which the company defaults in year $t+1$ (respective of which agency rates that instrument); the variable takes a value of zero otherwise. The other variables were defined in Table 1. Panel C reports summary statistics for the default sample, split along the investment-grade threshold.

Panel A								
	AAA	AA	A	BBB	BB	B	C	Total
Default in $t+1=0$	899	2219	3037	5796	4322	1592	136	18,001
Default in $t+1=1$	0	0	25	134	248	248	51	706
Total	899	2219	3062	5930	4570	1840	187	18,707
Panel B								
	Obs.	Mean	Std. dev.	Min.	Max.			
Default in $t+1$	18,707	0.038	0.191	0.000	1.000			
Issuer rating	18,707	8.823	3.888	1.000	18.000			
Non-rating services	18,707	0.053	0.225	0.000	1.000			
Ln(Non-rating issuer revenue)	18,508	0.027	0.200	0.000	3.792			
Panel C								
Investment-grade subsample (BBB- or higher)								
	Obs.	Mean	Std. dev.	Min.	Max.			
Default in $t+1$	12,110	0.013	0.114	0.000	1.000			
Issuer rating	12,110	6.661	2.914	1.000	10.429			
Non-rating services	12,110	0.079	0.269	0.000	1.000			
Ln(Non-rating issuer revenue)	11,925	0.041	0.247	0.000	3.792			
High yield subsample (BB+ or lower)								
	Obs.	Mean	Std. dev.	Min.	Max.			
Default in $t+1$	6597	0.083	0.276	0.000	1.000			
Issuer rating	6597	12.792	1.715	10.500	18.000			
Non-rating services	6597	0.007	0.081	0.000	1.000			
Ln(Non-rating issuer revenue)	6583	0.001	0.027	0.000	1.506			

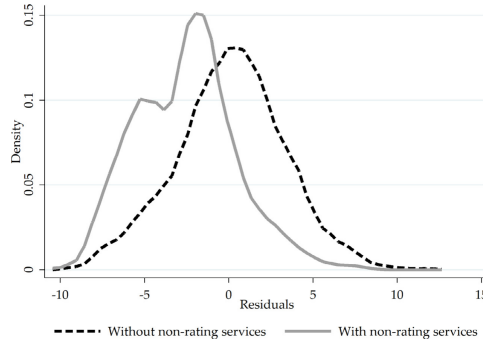


Fig. 1. Ratings of firms with and without non-rating services. This figure shows the distribution of issuer ratings for firms that obtain non-rating services and those that do not, after accounting for differences due to industry effects. Specifically, we plot the residuals from the following regression: $(\text{Issuer rating})_{i,j,t} = \gamma_{i-k} + \delta_{j,t} + \epsilon_{i,j,t}$, where i denotes the firm, k the industry, j denotes a rating agency, t denotes the year, and γ_{i-k} are industry dummies. The variables are defined in Table 1. The sample period is 2010–2015.

5.2 Figure 2, Table 3, and Table 4: main rating inflation tests

Read:

- Table 3 uses the non-rating-services dummy. Coefficients range from -0.281 to -0.326 in columns 1–3.
- In the saturated specification with issuer-year and agency-year fixed effects, the coefficient is -0.299. This means the agency with a consulting tie gives about a 0.3-notch better rating.
- Table 4 uses the log revenue measure. Coefficients range from -0.362 to -0.414.
- The authors interpret the magnitude as follows: a doubling of non-rating fees is associated with roughly a 0.4-notch better rating.
- Figure 2 shows the same intensive-margin pattern graphically: the fitted line slopes downward, so larger fees are associated with better rating differences.

Economic message: The same issuer in the same year receives a better rating from the agency it pays for non-rating services. This is the central rating-inflation result.

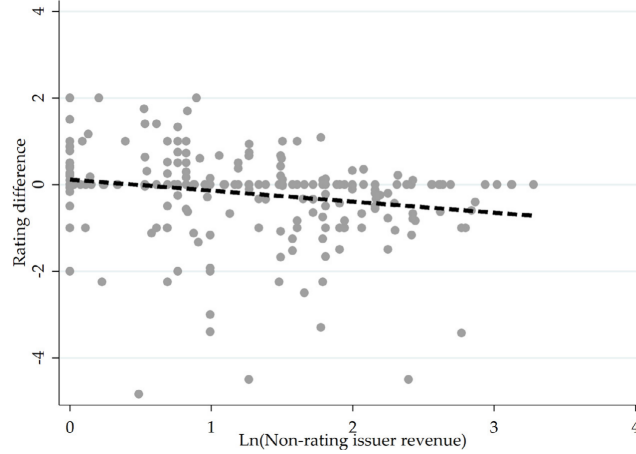


Fig. 2. Non-rating revenue and ratings. This figure plots the *Rating difference* (the difference between the *Issuer rating* from one rating agency minus the cross-sectional average of the ratings obtained from the other agencies in that year) against the variable $\text{Ln}(\text{Non-rating issuer revenue})$ and fits a linear prediction plot. *Issuer rating* and $\text{Ln}(\text{Non-rating issuer revenue})$ are defined in Table 1. The sample underlying the figure consists of observations in which issuers have multiple ratings and where those issuers pay an agency for consulting (i.e., where the variable *Non-rating services* takes the value of one). The sample period is 2010–2015.

Table 3

Ratings and the provision of non-rating services.

This table reports the coefficients for linear regression models estimating the association between ratings and the provision of non-rating services. Each observation corresponds to an issuer-agency-year. The variables are defined in Table 1. The sample period is 2010–2015. Heteroskedasticity-robust standard errors, clustered by issuer, are reported below coefficients. * denotes estimates that are significantly different from zero at the 10% level, ** at the 5% level, and *** at the 1% level.

	(1)	(2)	(3)
	Issuer rating		
Non-rating services	−0.281*** (0.058)	−0.326*** (0.068)	−0.299*** (0.068)
Issuer F.E.	x		
Year F.E.	x		
Agency F.E.	x	x	
Issuer x Year F.E.		x	x
Agency x Year F.E.			x
Observations	26,760	26,760	26,760
Adjusted R-squared	0.937	0.941	0.943

Table 4

Ratings and payments for non-rating services.

This table reports the coefficients for linear regression models estimating the association between ratings and revenue from issuers. Each observation corresponds to an issuer-agency-year. The variables are defined in Table 1. The sample period is 2010–2015. Heteroskedasticity-robust standard errors, clustered by issuer, are reported below coefficients. * denotes estimates that are significantly different from zero at the 10% level, ** at the 5% level, and *** at the 1% level.

	(1)	(2)	(3)
	Issuer rating		
$\text{Ln}(\text{Non-rating issuer revenue})$	−0.414*** (0.058)	−0.404*** (0.063)	−0.362*** (0.065)
Issuer F.E.	x		
Year F.E.	x		
Agency F.E.	x	x	
Issuer x Year F.E.		x	x
Agency x Year F.E.			x
Observations	26,468	26,468	26,468
Adjusted R-squared	0.936	0.938	0.939

5.3 Table 5 and Table 6: lagged payments and threshold incentives

Read:

- Table 5 adds lagged non-rating revenue. Lagged payments remain negative and significant in all specifications.
- This suggests that the commercial relationship is not purely contemporaneous; past payments may also matter for current ratings.
- Table 6 focuses on issuers close to important thresholds, especially AA- and BBB-.
- In Panel A, the dummy coefficients are about -0.742 to -0.927, and revenue coefficients are about -1.020 to -1.176.
- In Panel B, restricting to issuers with split ratings near thresholds makes effects even larger: dummy coefficients are about -1.024 to -1.209 and revenue coefficients are about -1.043 to -1.300.

Economic message: The conflict is strongest exactly where the issuer has the most to gain from one more notch. This is a powerful economic validation check.

Table 5

Ratings and past payments for non-rating services.

This table reports the coefficients for linear regression models estimating the association between ratings and non-rating revenue from issuers. Each observation corresponds to an issuer-agency-year. $\text{Lag} /$ is the lag operator and denotes one-year lags of the relevant variable. The variables are defined in Table 1. The sample period is 2010–2015. Heteroskedasticity-robust standard errors, clustered by issuer, are reported below coefficients. * denotes estimates that are significantly different from zero at the 10% level, ** at the 5% level, and *** at the 1% level.

	(1)	(2)	(3)
	Issuer rating		
Ln(Non-rating issuer revenue)	−0.255*** (0.065)	−0.164* (0.090)	−0.120 (0.089)
Lag[Ln(Non-rating issuer revenue)]	−0.223*** (0.056)	−0.198** (0.087)	−0.215*** (0.081)
Issuer F.E.	x		
Year F.E.	x		
Agency F.E.	x	x	
Issuer x Year F.E.		x	x
Agency x Year F.E.			x
Observations	16,811	16,811	16,811
Adjusted R-squared	0.950	0.954	0.954

Table 6

Ratings and non-rating services: subsamples for issuers close to important thresholds in the ratings spectrum.

This table reports the coefficients for linear regression models estimating the association between ratings and the purchase of non-rating services by issuers. Each observation corresponds to an issuer-agency-year. The variables are defined in Table 1. The sample period is 2010–2015. These tests focus on issuers that are close to an important “threshold” on the rating scale (AA- or BBB-), which is used for contracting and regulatory purposes (see Section 2.2). Specifically, in Panel A, the sample contains issuers that in a given year obtain at least one of the following “close” ratings from a rating agency that they do not pay for consulting: A+, A, A-, BB+, BB, or BB-. In Panel B, the tests focus on issuers that (i) in a given year obtain at least one “close” rating (A+, A, A-, BB+, BB, or BB-) from a rating agency that they do not pay for consulting, and that (ii) have at least two ratings in that year that are different from each other. Heteroskedasticity-robust standard errors, clustered by issuer, are reported below coefficients. * denotes estimates that are significantly different from zero at the 10% level, ** at the 5% level, and *** at the 1% level.

Panel A: issuers close to AA- or BBB- thresholds						
	(1)	(2)	(3)	(4)	(5)	(6)
Dependent variable:	Issuer rating					
Non-rating services	−0.742*** (0.258)	−0.927*** (0.232)	−0.848*** (0.221)			
Ln(Non-rating issuer revenue)				−1.084*** (0.169)	−1.176*** (0.146)	−1.020*** (0.175)
Issuer F.E.	x			x		
Year F.E.	x			x		
Agency F.E.	x	x		x	x	
Issuer x Year F.E.		x	x		x	x
Agency x Year F.E.			x			x
Observations	11,582	11,582	11,582	11,561	11,561	11,561
Adjusted R-squared	0.934	0.871	0.874	0.934	0.871	0.873
Panel B: issuers with split ratings close to AA- or BBB- thresholds						
	(1)	(2)	(3)	(4)	(5)	(6)
Dependent variable:	Issuer rating					
Non-rating services	−1.125*** (0.313)	−1.209*** (0.271)	−1.024*** (0.269)			
Ln(Non-rating issuer revenue)				−1.286*** (0.148)	−1.300*** (0.136)	−1.043*** (0.197)
Issuer F.E.	x			x		
Year F.E.	x			x		
Agency F.E.	x	x		x	x	
Issuer x Year F.E.		x	x		x	x
Agency x Year F.E.			x			x
Observations	1765	1765	1765	1752	1752	1752
Adjusted R-squared	0.861	0.855	0.861	0.860	0.854	0.859

5.4 Table 7 and Table 8: robustness to financial-sector issuers and ordered probit

Read:

- Table 7 shows that the main rating result is not driven only by financial-services issuers.
- Excluding financial-services firms, non-rating services remain associated with better ratings: dummy coefficients are about -0.150 to -0.249 and revenue coefficients are about -0.264 to -0.310.
- Within financial-services firms alone, the revenue measure remains strongly negative and significant.
- Table 8 uses ordered probit models rather than OLS.
- Ordered probit estimates remain negative and significant: about -0.592 to -0.616 for the dummy and -0.752 to -0.970 for the revenue measure.

Economic message: The favorable-rating result survives alternative samples and nonlinear rating models, so it is not an artifact of OLS or of the banking/financial sector.

Table 7

Ratings and non-rating services: issuers in the financial services sector versus issuers in other sectors.

This table reports the coefficients for linear regression models estimating the association between ratings and the purchase of non-rating services by issuers. Each observation corresponds to an issuer-agency-year. The variables are defined in Table 1. The sample period is 2010–2015. The sample in Panel A excludes financial services firms, while the sample in Panel B focuses on a sample of only financial services firms. Heteroskedasticity-robust standard errors, clustered by issuer, are reported below coefficients. * denotes estimates that are significantly different from zero at the 10% level, ** at the 5% level, and *** at the 1% level.

Panel A: excluding issuers in financial services						
	(1)	(2)	(3)	(4)	(5)	(6)
	Issuer rating					
Non-rating services	–0.150** (0.072)	–0.249*** (0.083)	–0.247*** (0.083)			
Ln(Non-rating issuer revenue)				–0.264*** (0.094)	–0.310*** (0.114)	–0.277*** (0.114)
Issuer F.E.	x			x		
Year F.E.	x			x		
Agency F.E.	x	x		x	x	
Issuer x Year F.E.			x			x
Agency x Year F.E.			x			x
Observations	24,004	24,004	24,004	23,859	23,859	23,859
Adjusted R-squared	0.926	0.928	0.929	0.926	0.926	0.927

Panel B: only issuers in financial services						
	(1)	(2)	(3)	(4)	(5)	(6)
	Issuer rating					
Non-rating services	–0.320*** (0.088)	–0.075 (0.105)	–0.078 (0.102)			
Ln(Non-rating issuer revenue)				–0.370*** (0.077)	–0.272*** (0.084)	–0.257*** (0.088)
Issuer F.E.	x			x		
Year F.E.	x			x		
Agency F.E.	x	x		x	x	
Issuer x Year F.E.		x	x		x	x
Agency x Year F.E.			x			x
Observations	2756	2756	2756	2609	2609	2609
Adjusted R-squared	0.961	0.963	0.965	0.960	0.961	0.962

Table 8

Ratings and non-rating services—ordered probit regressions.

This table reports the coefficients of ordered probit regression models estimating the association between ratings and the provision of non-rating services in Panel A, and between ratings and consulting revenue from issuers in Panel B. Each observation corresponds to an issuer-agency-year. In these tests, the dependent variable *Issuer rating* is rounded to whole numbers. The variables are defined in Table 1. The sample period is 2010–2015. This subsample only consists of observations in which issuers have multiple raters in a given year (see Table 1, Panel B). Heteroskedasticity-robust standard errors, clustered by issuer, are reported below coefficients. * denotes estimates that are significantly different from zero at the 10% level, ** at the 5% level, and *** at the 1% level.

Panel A: ordered probit specifications corresponding to Table 3			
	(1)	(2)	(3)
	Issuer rating		
Non-rating services	–0.592*** (0.108)	–0.616*** (0.151)	–0.577*** (0.156)
Issuer F.E.	x		
Year F.E.	x		
Agency F.E.	x	x	
Issuer x Year F.E.		x	x
Agency x Year F.E.			x
Observations	5141	5141	5141
Pseudo R-squared	0.584	0.649	0.657

Panel B: Ordered probit specifications corresponding to Table 4			
	(1)	(2)	(3)
	Issuer rating		
Ln(Non-rating issuer revenue)	–0.752*** (0.093)	–0.970*** (0.154)	–0.933*** (0.164)
Issuer F.E.	x		
Year F.E.	x		
Agency F.E.	x	x	
Issuer x Year F.E.		x	x
Agency x Year F.E.			x
Observations	4979	4979	4979
Pseudo R-squared	0.582	0.649	0.657

5.5 Figure 3, Figure 4, Table 9, and Table 10: default evidence

Read:

- Figure 3 confirms that default rates rise sharply as ratings deteriorate.
- Figure 4 compares default rates within rating categories for firms with and without non-rating-services relationships.
- Within several rating categories, firms paying for non-rating services default more often.
- Table 9 shows this in regressions. The non-rating-services coefficient is positive in every specification, ranging from 0.011 to 0.034.
- In columns 5–8 of Table 9, the model controls for rating-notch fixed effects. The result remains positive, so consulting clients default more even within rating categories.
- Table 10 splits the sample. In investment grade, the coefficient is around 0.005–0.008. In high yield, it is much larger, about 0.107–0.134.

Economic message: Better ratings for consulting clients are not justified by lower default risk. Within rating categories, these issuers default more, which directly points to lower rating quality.

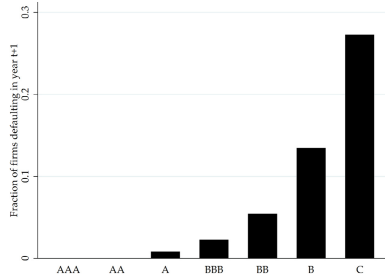


Fig. 3. Ratings and default rates. The figure shows one-year default rates by rating category. Observations are divided into coarse issuer rating categories. For each of the rating categories, the fraction of firms that default in the following year is shown on the vertical axis. The sample period is 2010–2014 (using default information until September 2015).

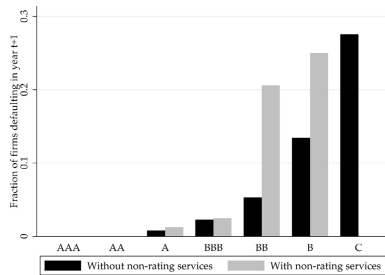


Fig. 4. Ratings and default rates: the role of non-rating services. The figure shows one-year default rates by rating category; for each rating category, default rates are separately reported for firms that pay for non-rating services and those that do not. The sample period is 2010–2014 (using default information until September 2015).

Table 9

Ratings, defaults, and the provision of non-rating services.

This table reports the coefficients for linear regression models estimating the association between default rates and the provision of non-rating services. Each observation corresponds to an issuer-agency-year. The variables are defined in Tables 1 and 2. In columns 5 to 8, *Issuer rating* is rounded to whole numbers and one dummy variable per rating notch is included. The sample period is 2010–2014 (using default information until September 2015). Heteroskedasticity-robust standard errors, clustered by issuer, are reported below coefficients. * denotes estimates that are significantly different from zero at the 10% level, ** at the 5% level, and *** at the 1% level.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Default in $t+1$							
Non-rating services	0.027*** (0.004)	0.034*** (0.005)	0.027*** (0.005)	0.029*** (0.005)	0.015*** (0.005)	0.014*** (0.005)	0.015*** (0.005)	0.011*** (0.005)
Issuer rating	0.010*** (0.000)	0.010*** (0.000)	0.011*** (0.001)	0.011*** (0.001)				
Constant	-0.053*** (0.003)							
Issuer rating F.E.								
Agency F.E.		x	x		x	x	x	
Year F.E.			x			x		
Industry F.E.			x			x		
Industry x Year F.E.				x			x	x
Agency x Year F.E.							x	
Agency x Issuer rating x Year F.E.								x
Observations	18,707	18,707	18,707	18,707	18,707	18,707	18,707	18,707
Adjusted R-squared	0.039	0.042	0.054	0.056	0.062	0.073	0.076	0.103

Table 10

Ratings, defaults, and the provision of non-rating services: investment-grade versus high yield firms.

This table reports the coefficients for linear regression models estimating the association between default rates and the provision of non-rating services. Each observation corresponds to an issuer-agency-year. The variables are defined in Tables 1 and 2. Panel A shows results for the sample of investment-grade firms, while Panel B reports results for the high yield subsample. In columns 5 to 8, *Issuer rating* is rounded to whole numbers and one dummy variable per rating notch is included. The sample period is 2010–2014 (using default information until September 2015). Heteroskedasticity-robust standard errors, clustered by issuer, are reported below coefficients. * denotes estimates that are significantly different from zero at the 10% level, ** at the 5% level, and *** at the 1% level.

Panel A: investment grade								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Default in $t+1$							
Non-rating services	0.003 (0.003)	0.007*** (0.003)	0.008*** (0.003)	0.008*** (0.003)	0.006*** (0.003)	0.006*** (0.003)	0.007*** (0.003)	0.005 (0.003)
Issuer rating	0.003*** (0.000)	0.004*** (0.000)	0.003*** (0.000)	0.003*** (0.000)				
Constant	-0.010*** (0.002)							
Issuer rating F.E.								
Agency F.E.		x	x		x	x	x	
Year F.E.		x	x			x		
Industry F.E.			x			x		
Industry x Year F.E.				x			x	x
Agency x Year F.E.							x	
Agency x Issuer rating x Year F.E.								x
Observations	12,110	12,110	12,110	12,110	12,110	12,110	12,110	12,110
Adjusted R-squared	0.007	0.009	0.015	0.016	0.010	0.015	0.017	0.022
Panel B: high yield								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Default in $t+1$							
Non-rating services	0.132*** (0.064)	0.134*** (0.065)	0.112*** (0.064)	0.129*** (0.065)	0.129*** (0.065)	0.107*** (0.064)	0.124*** (0.065)	0.129*** (0.065)
Issuer rating	0.028*** (0.003)	0.030*** (0.003)	0.029*** (0.003)	0.029*** (0.003)				
Constant	-0.282*** (0.033)							
Issuer rating F.E.								
Agency F.E.		x	x		x	x	x	
Year F.E.			x			x		
Industry F.E.			x			x		
Industry x Year F.E.				x			x	x
Agency x Year F.E.							x	
Agency x Issuer rating x Year F.E.								x
Observations	6597	6597	6597	6597	6597	6597	6597	6597
Adjusted R-squared	0.032	0.037	0.069	0.087	0.043	0.073	0.090	0.114

5.6 Table 11 and Table 12: fee amounts and default rates

Table 11

Ratings, defaults, and payments for non-rating services.

This table reports the coefficients for linear regression models estimating the association between default rates and payments for non-rating services. Each observation corresponds to an issuer-agency-year. The variables are defined in Tables 1 and 2. In columns 5 to 8, *Issuer rating* is rounded to whole numbers and one dummy variable per rating notch is included. The sample period is 2010–2014 (using default information until September 2015). Heteroskedasticity-robust standard errors, clustered by issuer, are reported below coefficients. * denotes estimates that are significantly different from zero at the 10% level, ** at the 5% level, and *** at the 1% level.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Default in $t+1$							
Ln(Non-rating issuer revenue)	0.020*** (0.003)	0.026*** (0.003)	0.018*** (0.003)	0.019*** (0.003)	0.008*** (0.002)	0.008*** (0.003)	0.008*** (0.003)	0.003 (0.003)
Issuer rating	0.019*** (0.000)	0.010*** (0.000)	0.011*** (0.001)	0.011*** (0.001)				
Constant	−0.051*** (0.003)							
Issuer rating F.E.					x	x	x	
Agency F.E.					x	x		
Year F.E.		x	x		x	x		
Industry F.E.			x			x		
Industry x Year F.E.				x			x	x
Agency x Year F.E.				x			x	
Agency x Issuer rating x Year F.E.								x
Observations	18,508	18,508	18,508	18,508	18,508	18,508	18,508	18,508
Adjusted R-squared	0.039	0.041	0.053	0.055	0.062	0.073	0.076	0.103

Table 12

Ratings, defaults, and payments for non-rating services: investment-grade versus high yield firms.

This table reports the coefficients for linear regression models estimating the association between default rates and payments for non-rating services. Each observation corresponds to an issuer-agency-year. The variables are defined in Tables 1 and 2. Panel A shows results for the sample of investment-grade firms, while Panel B reports results for the high yield subsample. In columns 5 to 8, *Issuer rating* is rounded to whole numbers and one dummy variable per rating notch is included. The sample period is 2010–2014 (using default information until September 2015). Heteroskedasticity-robust standard errors, clustered by issuer, are reported below coefficients. * denotes estimates that are significantly different from zero at the 10% level, ** at the 5% level, and *** at the 1% level.

<i>Panel A: investment grade</i>								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Default in $t+1$							
Ln(Non-rating issuer revenue)	0.003 (0.002)	0.006*** (0.002)	0.007*** (0.002)	0.007*** (0.003)	0.004** (0.002)	0.005** (0.002)	0.006** (0.003)	0.002 (0.003)
Issuer rating	0.003*** (0.000)	0.004*** (0.000)	0.003*** (0.000)	0.003*** (0.000)				
Constant	−0.010*** (0.002)							
Issuer rating F.E.					x	x	x	
Agency F.E.		x	x		x	x		
Year F.E.			x		x	x		
Industry F.E.			x			x		
Industry x Year F.E.				x			x	x
Agency x Year F.E.				x			x	
Agency x Issuer rating x Year F.E.								x
Observations	11,925	11,925	11,925	11,925	11,925	11,925	11,925	11,925
Adjusted R-squared	0.007	0.009	0.015	0.016	0.009	0.015	0.016	0.022
<i>Panel B: high yield</i>								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Default in $t+1$							
Ln(Non-rating issuer revenue)	0.149 (0.172)	0.164 (0.174)	0.102 (0.169)	0.194 (0.193)	0.166 (0.179)	0.108 (0.173)	0.201 (0.194)	0.197 (0.191)
Issuer rating	0.028*** (0.003)	0.030*** (0.003)	0.029*** (0.003)	0.029*** (0.003)				
Constant	−0.281*** (0.033)							
Issuer rating F.E.					x	x	x	
Agency F.E.		x	x		x	x		
Year F.E.			x		x	x		
Industry F.E.			x			x		
Industry x Year F.E.				x			x	x
Agency x Year F.E.				x			x	
Agency x Issuer rating x Year F.E.								x
Observations	6583	6583	6583	6583	6583	6583	6583	6583
Adjusted R-squared	0.031	0.036	0.068	0.086	0.042	0.072	0.090	0.113

Read:

- Table 11 replaces the dummy with the log amount of non-rating revenue.
- The coefficient is positive and significant in most specifications, suggesting that higher fees are associated with higher default risk.
- In the most saturated column with agency-by-rating-by-year fixed effects, the coefficient falls to 0.003 and is not statistically significant.
- Table 12 splits by investment grade and high yield.
- Investment-grade coefficients are small but often significant, around 0.004–0.007 in several columns.
- High-yield coefficients are economically large but imprecise and not statistically significant.

Economic message: The intensive-margin default evidence is somewhat weaker than the ratings evidence, but it points in the same direction: larger non-rating payments do not correspond to safer borrowers.

6 Short summary

- The paper studies whether rating agencies' non-rating revenues create conflicts of interest.
- The setting is India after SEBI required rating agencies to disclose issuer-level non-rating income.

- The main data unit is issuer-agency-year.
- The paper compares ratings of the same issuer in the same year across different rating agencies.
- Rating agencies give better ratings when they also receive non-rating revenue from the issuer.
- The better-rating effect is about 0.3 notches for the services dummy and about 0.4 notches for a doubling of fees.
- The effect is larger near important ratings thresholds such as AA- and BBB-.
- Within rating categories, consulting clients default more often.
- The default evidence rejects the benign interpretation that non-rating services merely reveal lower credit risk.
- The results support the view that commercial ties can compromise the quality of issuer-paid ratings.

7 Conclusion

7.1 Core conclusion

Agencies that earn non-rating revenue from an issuer assign that issuer more favorable ratings than other agencies rating the same issuer in the same year. These favorable ratings are not matched by better ex post credit performance; instead, the issuers default more often within rating categories.

7.2 Economic intuition

The issuer-pays model gives issuers a channel to reward rating agencies. Non-rating services make that channel more flexible because the size and scope of consulting revenue can vary. The agency therefore has a stronger incentive to maintain a profitable client relationship, especially around thresholds where the issuer gains substantially from one more notch.

7.3 Takeaway

The paper's central message is not simply that consulting exists. It is that the flow of non-rating payments is associated with optimistic ratings and higher default rates. This means the entire commercial relationship between issuers and raters matters for rating quality.